

LAPORAN PRAKTIKUM
ADMINISTRASI DAN MANAJEMEN JARINGAN

UNIT 6
DNS SERVER



Disusun oleh:

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SEKOLAH VOKASI
UNIVERSITAS GADJAH MADA

2026

UNIT 6

DNS SERVER

A. Tujuan Praktikum

- Mahasiswa mampu melakukan instalasi DNS Server
- Mahasiswa mampu memodifikasi konfigurasi DNS Server sehingga alamat IP server dapat diakses menggunakan nama domain tertentu.

B. Dasar Teori DNS Server

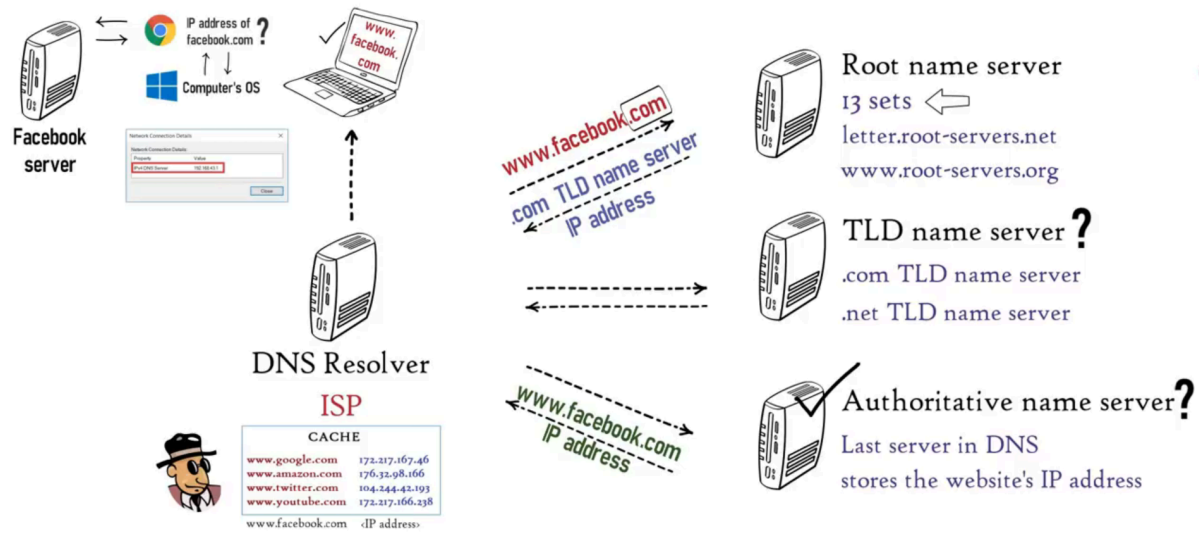
DNS (Domain Name System) merupakan sistem berbentuk database terdistribusi yang akan memetakan/ mengkonversikan nama host/mesin/domain ke alamat IP (Internet Protocol) dan sebaliknya dari alamat IP ke nama host yang disebut dengan reverse-mapping.

DNS berperan sebagai jembatan, yaitu karena:

- Manusia lebih mudah untuk mengingat nama daripada alamat IP dengan panjang 32 bit.
- Komputer menggunakan alamat IP untuk berkomunikasi dan berinteraksi.

Cara kerja DNS adalah sebagai berikut:

- Proses kerja DNS dimulai ketika pengguna memasukkan URL atau nama domain ke dalam browser mereka. Browser akan meminta informasi dari server DNS lokal yang terhubung ke jaringan. Jika informasi yang diminta tidak tersedia pada server lokal, permintaan akan diteruskan ke server DNS lainnya dalam hierarki DNS.
- Server DNS akan mencari informasi yang sesuai dengan nama domain yang diminta dan mengembalikan alamat IP yang terkait dengan nama domain tersebut. Informasi ini kemudian dikirim kembali ke browser, sehingga browser dapat memulai koneksi ke server yang terkait.
- Proses ini melibatkan beberapa jenis server DNS yang berbeda, termasuk DNS resolver, Root name server, Top Level Domain (TLD) name server, dan Authoritative name server



C. Langkah Praktikum DNS Server

1. Masuk Ubuntu Server anda kemudian cek alamat IP Ubuntu di virtual box dengan mengetik:

ip a

```
root@sultan:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d0:31:7b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 81186sec preferred_lft 81186sec
    inet6 fd17:625c:f037:2:a00:27ff:fed0:317b/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86203sec preferred_lft 14203sec
    inet6 fe80::a00:27ff:fed0:317b/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a0:97:06 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.107/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 490sec preferred_lft 490sec
    inet6 fe80::a00:27ff:fea0:9706/64 scope link
        valid_lft forever preferred_lft forever
root@sultan:~#
```

Catat IP address tersebut kemudian bukalah Command Prompt (CMD) dari Windows. Kemudian coba ping alamat IP Ubuntu virtual box. Pastikan ping mendapat balasan (Reply).

```
C:\Users\TAJ>ping 192.168.56.107

Pinging 192.168.56.107 with 32 bytes of data:
Reply from 192.168.56.107: bytes=32 time<1ms TTL=64
Reply from 192.168.56.107: bytes=32 time=1ms TTL=64
Reply from 192.168.56.107: bytes=32 time=1ms TTL=64
Reply from 192.168.56.107: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

2. Pikirkan nama domain yang akan anda gunakan, misalkan (namadomain.com). Cobalah ping nama domain tersebut melalui CMD Windows. Pastikan anda tidak mendapat balasan atau muncul keterangan: Ping request could not find host namadomain.com. Please check the name and try again..

```
C:\Users\TAJ>ping ubuntu Sultan.com
Ping request could not find host ubuntu Sultan.com. Please check the name and try again.
```

3. Kembali ke Ubuntu server dan login root. Update ubuntu dengan apt update.

```
root@sultan:/# apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@sultan:/#
```

Install aplikasi DNS yaitu Bind9 dengan mengetik:

```
apt-get install bind9
```

```
root@sultan:/# apt-get install bind9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-utils dns-root-data
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-utils dns-root-data
0 upgraded, 3 newly installed, 0 to remove and 91 not upgraded.
Need to get 423 kB of archives.
After this operation, 1,637 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Cek hasil instalasi dengan mengetik

```
named -v
```

```
systemctl status named
```

```
root@sultan:/# named -v
BIND 9.18.39-0ubuntu0.24.04.7-Ubuntu (Extended Support Version) <id:>
root@sultan:/# systemctl status named
• named.service - BIND Domain Name Server
  Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
  Active: active (running) since Fri 2026-10-02 02:41:05 UTC; 18s ago
    Docs: man:named(8)
  Main PID: 3426 (named)
    Status: "running"
     Tasks: 6 (limit: 9204)
  Memory: 23.7M (peak: 24.2M)
     CPU: 76ms
  CGroup: /system.slice/named.service
          └─3426 /usr/sbin/named -f -u bind

Oct 02 02:41:05 sultan named[3426]: managed-keys-zone: loaded serial 0
Oct 02 02:41:05 sultan named[3426]: zone localhost/IN: loaded serial 2
Oct 02 02:41:05 sultan named[3426]: zone 0.in-addr.arpa/IN: loaded serial 1
Oct 02 02:41:05 sultan named[3426]: zone 127.in-addr.arpa/IN: loaded serial 1
Oct 02 02:41:05 sultan named[3426]: zone 255.in-addr.arpa/IN: loaded serial 1
Oct 02 02:41:05 sultan named[3426]: all zones loaded
Oct 02 02:41:05 sultan named[3426]: running
Oct 02 02:41:05 sultan systemd[1]: Started named.service - BIND Domain Name Server.
Oct 02 02:41:15 sultan named[3426]: resolver priming query complete: timed out
```

Selanjutnya update rule firewall Ubuntu agar mengizinkan bind9 dengan megetik
ufw allow bind9

```
root@sultan:/# ufw allow bind9
Rule added
Rule added (v6)
root@sultan:/#
```

4. Lakukan konfigurasi dengan mengetik perintah berikut.

```
cd /etc/bind/
```

```
nano named.conf.options
```

hilangkan tanda double slash di bagian forwarders dan ketikkan alamat IP yang anda gunakan

```

GNU nano 7.2                                named.conf.options
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.
    forwarders {
        192.168.56.107;
    };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};

```

Simpan dan keluar nano.

5. Konfigurasi bind9

nano named.conf.local

tambahkan script berikut di bagian akhir file:

```

zone "namadomain.com" {
    type master;
    file "/etc/bind/db.namadomain1";
};

zone "56.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.namadomain2";
};

```

Setelah itu simpan dan keluar dari nano editor.

```

GNU nano 7.2                                named.conf.local *
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "ubuntusultan.com" {
type master;
file "/etc/bind/db.ubuntusultan1";
};

zone "56.168.192.in-addr.arpa" {
type master;
file "/etc/bind/db.ubuntusultan2";
};

```

6. Salin file db.local dan db.127 dan salin dengan nama sesuai dengan nama server

```
cp db.local db.namadomain1
```

```
cp db.127 db.namadomain2
```

```

root@sultan:/etc/bind# cp db.local db.ubuntusultan1
root@sultan:/etc/bind# cp db.127 db.ubuntusultan2

```

7. Cek hasil salinan file dengan menggunakan perintah ls.

```

root@sultan:/etc/bind# ls
bind.keys  db.127  db.empty  db.ubuntusultan1  named.conf          named.conf.local  rndc.key
db.0       db.255  db.local  db.ubuntusultan2  named.conf.default-zones  named.conf.options  zones.rfc1918
root@sultan:/etc/bind#

```

8. Edit file **db.namadomain1** dengan mengetik

```
nano db.namadomain1
```

kemudian ganti sebagian isinya menjadi seperti berikut

```

@      IN      SOA      namadomain.com.    root.namadomain.com. (
                                2
                                ; Serial
                                604800
                                ; Refresh
                                86400
                                ; Retry
                                2419200
                                ; Expire
                                604800 )
                                ; Negative Cache TTL
;
@      IN      NS       namadomain.com.
@      IN      A        192.168.56.107
www    IN      A        192.168.56.107

```

Selanjutnya simpan hasil edit dan exit nano editor.

```

GNU nano 7.2                                     db.ubuntusultan1 *
;
; BIND data file for local loopback interface
;
$TTL      604800
@         IN      SOA      ubuntusultan.com. root.ubuntusultan.com. (
                                2          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ubuntusultan.com.
@         IN      A        192.168.56.107
www       IN      A        192.168.56.107

```

9. Edit file **db.namadomain2** dengan mengetik

nano db.**namadomain2**

kemudian ganti sebagian isinya menjadi seperti berikut

```

@         IN      SOA      namadomain.com.   root.namadomain.com. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       namadomain.com.
107      IN      PTR      namadomain.com.

```

Selanjutnya simpan hasil edit dan exit nano editor.

```

GNU nano 7.2                                     db.ubuntusultan2 *
;
; BIND reverse data file for local loopback interface
;
$TTL      604800
@         IN      SOA      ubuntusultan.com. root.ubuntusultan.com. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ubuntusultan.com.
107      IN      PTR      ubuntusultan.com.

```

10. Cek apakah ada salah pengetikan pada script

named-checkconf

named-checkzone db.**namadomain1** db.**namadomain2**

Jika tidak muncul error, berarti konfigurasi sudah benar.

```

root@sultan:/etc/bind# named-checkconf
root@sultan:/etc/bind# named-checkzone db.ubuntusultan1 db.ubuntusultan2
zone db.ubuntusultan1/IN: loaded serial 1
OK
root@sultan:/etc/bind#

```

11. Buka file konfigurasi netplan

`nano /etc/netplan/01-netconf.yaml`

Tambahkan nameserver di adapter enp0s8 sehingga menjadi seperti berikut:

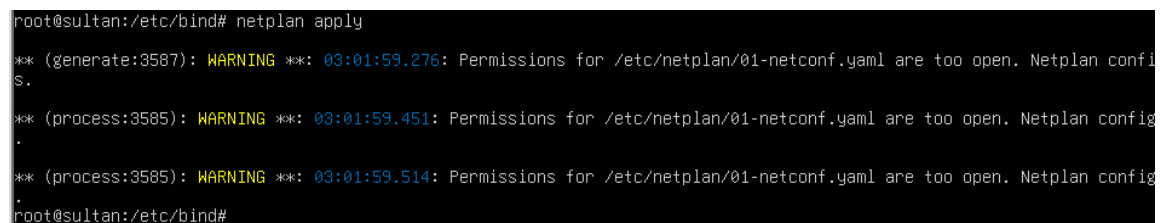
```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      dhcp4: yes
      nameservers:
        addresses:
          - 8.8.8.8
          - 192.168.56.107
```

A screenshot of the GNU nano 7.2 text editor editing the file /etc/netplan/01-netconf.yaml. The configuration is as follows:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      dhcp4: yes
      nameservers:
        addresses:
          - 8.8.8.8
          - 192.168.56.107
```

Simpan dan keluar file konfigurasi kemudian ketikkan

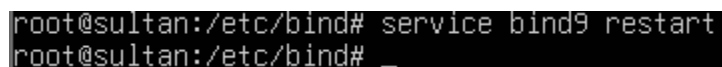
`netplan apply`

A screenshot of a terminal window showing the output of the 'netplan apply' command. The prompt is 'root@sultan:/etc/bind#'. The output consists of three warning messages from the systemd journal, all indicating that permissions for /etc/netplan/01-netconf.yaml are too open. The messages are:

```
root@sultan:/etc/bind# netplan apply
** (generate:3587): WARNING **: 03:01:59.276: Permissions for /etc/netplan/01-netconf.yaml are too open. Netplan config is not immutable.
** (process:3585): WARNING **: 03:01:59.451: Permissions for /etc/netplan/01-netconf.yaml are too open. Netplan config is not immutable.
** (process:3585): WARNING **: 03:01:59.514: Permissions for /etc/netplan/01-netconf.yaml are too open. Netplan config is not immutable.
root@sultan:/etc/bind#
```

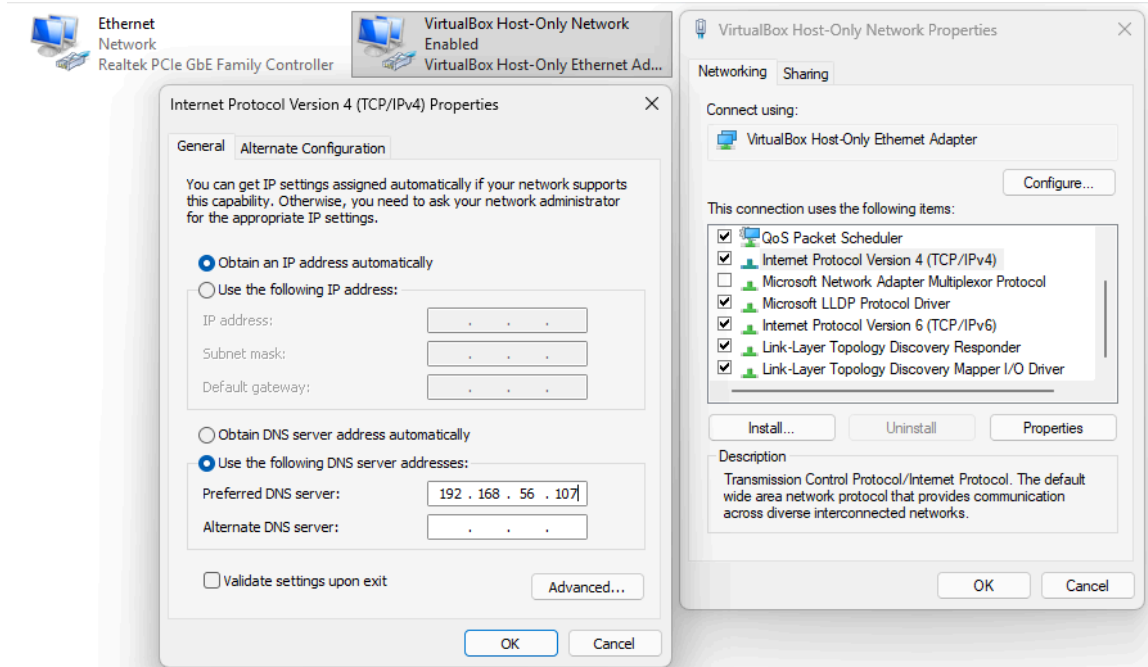
12. Restart aplikasi bind9

`service bind9 restart`

A screenshot of a terminal window showing the command 'service bind9 restart' being executed. The prompt is 'root@sultan:/etc/bind#'. The output shows the command being run and a cursor on the next line.

```
root@sultan:/etc/bind# service bind9 restart
root@sultan:/etc/bind#
```

13. Atur network adapter (VirtualBox Host-Only Ethernet Adapter) yang terhubung dengan VM anda agar menggunakan alamat DNS sesuai alamat IP Ubuntu Server.



D. Latihan

1. Test hasil konfigurasi DNS dengan perintah berikut:

nslookup namadomain.com

```
root@sultan:/etc/bind# nslookup ubuntusultan.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   ubuntusultan.com
Address: 192.168.56.107
```

nslookup 192.168.56.107

```
root@sultan:/etc/bind# nslookup 192.168.56.107
107.56.168.192.in-addr.arpa    name = ubuntusultan.com.

Authoritative answers can be found from:
```

dig namadomain.com

```

root@sultan:/etc/bind# dig ubuntusultan.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.7-Ubuntu <<>> ubuntusultan.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13840
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;ubuntusultan.com.                IN      A

;; ANSWER SECTION:
ubuntusultan.com.                7007    IN      A      192.168.56.107

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Oct 02 03:08:46 UTC 2026
;; MSG SIZE rcvd: 61

```

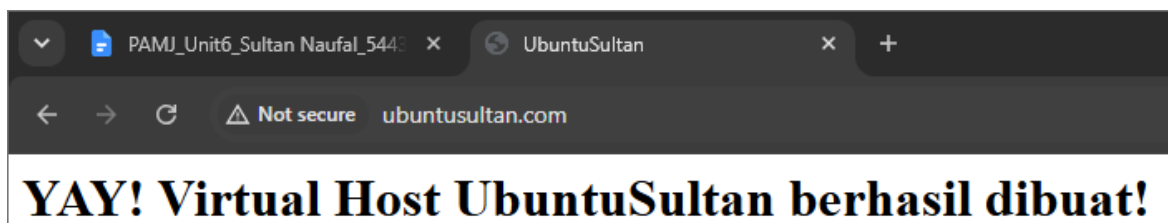
ping namadomain.com

```

root@sultan:/etc/bind# ping ubuntusultan.com
PING ubuntusultan.com (192.168.56.107) 56(84) bytes of data:
64 bytes from ubuntusultan.com (192.168.56.107): icmp_seq=1 ttl=64 time=0.014 ms
64 bytes from ubuntusultan.com (192.168.56.107): icmp_seq=2 ttl=64 time=0.065 ms
64 bytes from ubuntusultan.com (192.168.56.107): icmp_seq=3 ttl=64 time=0.087 ms
64 bytes from ubuntusultan.com (192.168.56.107): icmp_seq=4 ttl=64 time=0.052 ms
64 bytes from ubuntusultan.com (192.168.56.107): icmp_seq=5 ttl=64 time=0.048 ms

```

2. Tes dengan mengakses nama domain melalui browser di Windows!



3. Jelaskan fungsi perintah nslookup dan dig!

Nslookup digunakan untuk melakukan testing sederhana koneksi DNS

Dig digunakan sama seperti nslookup, yaitu mengecek koneksi DNS, namun dig memiliki output hasil yang lebih lengkap seperti QUESTION dan ANSWER.

E. Rekomendasi Bacaan

https://en.wikipedia.org/wiki/Domain_Name_System